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Keyboard, Mouse and Joystick Support of the G2 Amiga

USB keyboards, mice and Sony DualShock pads on the G2's hub, read by the RISC-V firmware's USB host and passed to the Amiga core's keyboard, mouse counters and two game ports. Two pads play as two Amiga joysticks; the same devices drive the setup screen.

HIGHLIGHTS

- Boot-protocol USB keyboard: every Amiga key, Help, both Amiga keys, keypad.
- Boot-protocol USB mouse on game port 1.
- Two DualShock 4 or DualShock 3 pads: pad 1 on the joystick port, pad 2 on the mouse port.
- Hot plug on every hub port, scanned every 2 s.
- F12 or PS opens setup; Ctrl-Amiga-Amiga resets.

KEY PARAMETERS

PARAMETER	VALUE
Keyboards	1, HID boot
Mice	1, HID boot
Pads	2, DualShock 3 / 4
Joystick bits	up, down, left, right, fire, fire 2
Stick threshold	±40 of 128
Scan period	2 s
Retry of a failed device	30 s

1 SUPPORTED DEVICES

CLASS	DEVICES	SPEED	AT MOST	PROTOCOL
Keyboard	Any USB keyboard with the HID boot protocol (subclass 1, protocol 1)	Low or full speed	1	SET_PROTOCOL boot, SET_IDLE; 8-byte boot reports on an interrupt endpoint
Mouse	Any USB mouse with the HID boot protocol (protocol 2)	Low or full speed	1	Boot reports: buttons, X, Y
Game pad	Sony DualShock 4 (054C:05C4, 054C:09CC), DualShock 4 USB wireless adapter (054C:0BA0), DualShock 3 (054C:0268)	Full speed, through the hub's transaction translator	2	HID reports of 64 bytes; DualShock 4: SET_IDLE and a report-descriptor request start the reports; DualShock 3: SET_REPORT feature 0xF4 = 42 0C 00 00

Every port of the G2 is behind its USB2514 hub; one port holds the stick. Devices are found by a scan of the hub ports every 2 s, opened in the order pad, keyboard, mouse, FTDI adapter, USB-MIDI interface, and released when unplugged; a device that fails to enumerate is retried after 30 s. A pad that fails 16 polls in a row is released. Keyboards and mice are usually low-speed (1.5 Mbit/s) devices: the firmware's USB driver runs them as split transactions through the hub's transaction translator.

2 GAME PADS AND THE GAME PORTS

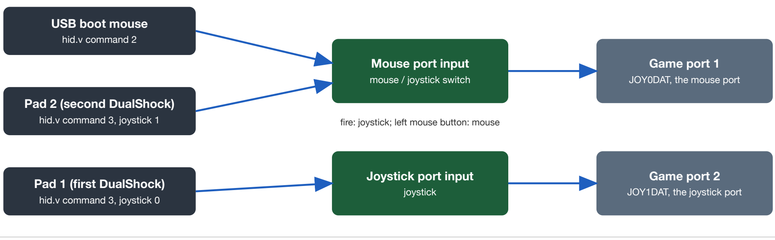


Figure 1. Input paths into the Amiga's two game ports. The first pad drives the joystick port, which single-player games read; the second pad shares the mouse port with the mouse, switched by the core as on an Amiga with a joystick and a mouse in the same socket.

PAD CONTROL	AMIGA INPUT	IN SETUP
D-pad, or the left stick past 40 of 128 from centre (below 88 or above 168 of 0–255)	Up, down, left, right	Item and page selection
Cross	Fire (button 1)	Enter
Circle	Fire 2 (button 2)	Escape
Start (OPTIONS on a DualShock 4)	—	Enter
Triangle	—	F10: Save and reset
PS	Opens setup	Leaves setup
Square, L1, R1, L2, R2, L3, R3, SELECT / SHARE, right stick, touch pad	Not used	Not used

PADS PRESENT	GAME PORT 2 (JOYSTICK)	GAME PORT 1 (MOUSE)
One	the pad	the mouse
Two	the first pad found	the mouse, or the second pad after its fire button is pressed (the left mouse button switches back)

A pad's state goes to the core as six bits {fire 2, fire, up, down, left, right} (hid.v command 3) whenever it changes; a pad unplugged with a direction or fire held is released in the core. The Peripherals page of setup shows each pad's hub port and game port.

3 KEYBOARD

PC KEY	AMIGA KEY
Letters, digits, punctuation, Return, Esc, Backspace, Tab, Space, Caps Lock	The key in the same position
F1–F10	F1–F10
F12	Opens setup (not sent to the Amiga)
F11, Print Screen, Scroll Lock, Pause, Num Lock	Not mapped
Delete	Del
End, Insert	Help
Home, Page Up	Keypad (and)
Page Down	Right Amiga
Left Windows / Command	Left Amiga
Right Windows / Command	Right Amiga
Left Alt, Right Alt	Left Alt, Right Alt
Left Ctrl	Ctrl
Left Shift, Right Shift	Left Shift, Right Shift
Cursor keys, keypad 0–9 . / * - + Enter	The same keys
Non-US # and Non-US \ keys (ISO keyboards)	The Amiga international key (code 0x2B)
Left Ctrl + Left Windows + Right Windows	Reset (Ctrl-Amiga-Amiga)

Each key and modifier change goes to the core (hid.v command 1) as {release, HID usage}; the core's key map turns the usage into the Amiga key code. The Amiga sees the keys as an Amiga keyboard reports them: the key map of AmigaOS (Prefs/Input) decides the characters. Right Ctrl is not mapped.

4 MOUSE

ITEM	VALUE
Report	Buttons, X and Y as signed 8-bit counts (boot protocol)
Path	hid.v command 2 {buttons, dx, dy} into the core's mouse counters of game port 1
Buttons	Left, right and middle
Wheel	Not used

5 HOT PLUG AND RECOVERY

EVENT	EFFECT
Device plugged in	Enumerated at the next scan (at most 2 s); the setup page lines change
Device unplugged	Released at the next scan; held pad state is released in the core
Third pad	Rejected: two pad slots
Unknown device	Rejected with its VID:PID in the boot log (\amiga\BOOT.LOG); retried after 30 s

6 FIRMWARE INTERFACE

STAGE	DETAIL
USB	Interrupt transfers on the EHCI periodic schedule; one slot each for the keyboard, the mouse and two pads; the endpoint's bInterval
Decode	DualShock reports to 17 button bits and four stick axes 0–255 (centre 128); boot reports to keys and mouse counts
Core	Bytes to the companion interface (MCU_TX: {start, target 1 = hid.v, byte}); hid.v command 1 keys, 2 mouse, 3 joystick {device, bits}
Setup	While setup is open the same inputs drive the setup screen and nothing goes to the core; the core is stopped

7 LIMITS AND UNTESTED ITEMS

ITEM	STATE
DualShock 4 and DualShock 3 on the hub	Tested on hardware: both pads enumerate and play on the joystick ports
Keyboard and mouse, low and full speed	Tested on hardware
Pad on the joystick port with games	Tested on hardware
Release of a held direction or fire when a pad is unplugged (firmware 1.0)	Built; the unplug case untested
Analogue sticks as proportional joysticks, CD32 pad buttons	Not supported
Other pads (Xbox, generic HID game pads)	Not supported: rejected as unknown devices

Sources: fw/amiga/amiga_main.c (input_poll, pads_scan, pad_send), fw/usbpad.c, fw/usbhid.c (not in the public source), amiga/nanomig/src/misc/hid.v and amiga_keymap.v. Repository mikewolak/pano-g2-amiga, release 1.0.